

Foundation (Jalapeno)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x + y = 2$
(A) $(x, y) = (0, 2)$
(B) $(x, y) = (2, 0)$
(C) $x = 1$, $y = 1$
(D) $x = -1$, $y = 3$
(E) No solution
- Q2.** What is the graphical representation of the system: $y = x^2$, $y = -x^2 + 2$?
(A) Two intersecting parabolas
(B) Two parallel parabolas
(C) One parabola and one line
(D) Two intersecting lines
(E) One line and one circle
- Q3.** Solve the system: $x^2 - 4y^2 = 1$, $x + y = 1$
(A) $x = 1$, $y = 0$
(B) $x = -1$, $y = 0$
(C) $x = 0$, $y = -1$
(D) $x = 2$, $y = -1$
(E) $x = -2$, $y = 1$
- Q4.** What is the solution to the system: $y = x^2 - 2x + 1$, $y = -(x - 2)^2 + 1$?
(A) $x = 0$, $y = 1$
(B) $x = 1$, $y = 0$
(C) $x = 2$, $y = 1$
(D) $x = -1$, $y = 4$
(E) $x = 2$, $y = -1$
- Q5.** The system $x^2 + y^2 = 4$ and $y = x^2 - 4$ has
(A) One solution
(B) Two solutions
(C) Three solutions
(D) Four solutions
(E) No solution
- Q6.** Solve the system: $x^2 + y^2 = 1$, $y = 2x$
(A) $x = 0$, $y = 0$
(B) $x = 1$, $y = 2$
(C) $x = -1$, $y = -2$
(D) $x = \frac{1}{\sqrt{5}}$, $y = \frac{2}{\sqrt{5}}$
(E) No solution
- Q7.** The system $y = x^2 - 2x + 1$ and $y = x^2 + 2x - 1$ has
(A) One solution
(B) Two solutions
(C) Three solutions
(D) Four solutions
(E) No solution
- Q8.** What is the solution to the system: $y = x^2 - 1$, $x + y = 2$?
(A) $x = -1$, $y = 0$
(B) $x = 1$, $y = 1$
(C) $x = -2$, $y = 3$
(D) $x = 2$, $y = -1$
(E) $x = 1$, $y = 0$

Open Ended Questions (FRQ)

Q9. Solve the system $x^2 + y^2 = 9$, $y = x - 3$. Show all work and explain your reasoning.

Q10. Graph the system $y = x^2$, $y = x^2 - 2x - 3$. Find the points of intersection and explain their significance.

Chef's Tips

- Check for any calculation errors
- Make sure to provide clear explanations for free response questions
- Consider providing a graph paper for questions that require graphing